

Rosacea, inflammation, and aging: The inefficiency of stress

Rosacea, inflammation, and aging: The inefficiency of stress Rosacea, or acne rosacea, has been defined as "vascular and follicular dilation involving the nose and contiguous portions of the cheeks . . ." that may involve persistent erythema with hyperplasia of sebaceous glands. Stedman's Medical Dictionary 23rd edition. Light-skinned people, especially women between the ages of 30 and 50, sometimes develop a persistent redness of their cheeks and nose. It may begin as a tendency to flush excessively, but the blood vessels can become chronically dilated. Similar processes occur in dark-skinned people less frequently. The eyes are sometimes involved, with redness of the exposed areas (conjunctival hyperemia). New blood vessels develop in the area, and the flow of blood through the affected tissue is greatly increased. The tissues become thickened and fibrotic, with the multiplication of fibroblasts and the increased deposition of collagen. The cornea normally receives its oxygen from the air, and its nutrients from the aqueous humor. As rosacea of the eye develops, the blood vessels surrounding the cornea become increasingly visible, and, especially on the inner (nasal) side of the eye, the vessels tend to enlarge and become tortuous. Rhinophyma, or potato nose, has been described as a late development of rosacea. Too often, the medical reaction is to give the condition a name, and to distinguish its variants as if they were different problems, and then to use the most direct means to eliminate the problem they have defined. A typical attitude is that "Rosacea is an enigmatic disease with multiple exacerbations and remissions, and, unfortunately, treatment is directed toward symptomatic control rather than cure" (Randleman). Lasers or other radiation, caustic chemical abrasion, surgical planing and dermal shaves, and other forms of surgery may be used to destroy the superficial blood vessels, and to reduce the enlarged nose or other irregularities. A few decades ago, when rosacea was believed to be the result of a local infection, antibiotics were used to treat it, and some of them, including tetracycline, helped. It was discovered that some antibiotics have anti-inflammatory actions, apart from their germicidal effects, and now it is very common to prescribe the chronic use of tetracycline to suppress symptoms. Rosacea, and the fibrotic changes associated with it (pingueculae and pterygia in the eyes, rhinophyma of the nose, etc.), are much more than "cosmetic" issues, involving the skin and eye surface. If the invasive proliferation of blood vessels can be prevented, it's important to do that, because, for example, pannus/neovascularization of the cornea can seriously impair vision. But possibly the strangest thing about the relationship of the medical profession to rosacea is that its essential features, invasive neovascularization and fibrotic growth, are of great interest when they occur elsewhere, and many physiological processes are known to regulate the growth of blood vessels and fibroblasts, but nearly all the attention given to rosacea and rhinophyma concerns control of symptoms for cosmetic effect. Rosacea is a physiological problem that deserves consideration in the light of all that's known about physiology and developmental biology. The increased incidence of rosacea after the age of 30, and the fact that it occurs most commonly in the areas that are most exposed to sunlight (bald men sometimes develop it on the top of the head), indicate that aging and irritation are essential causes. Stress, irritation (such as produced by ultraviolet or ionizing radiation or free radicals), and aging are known to cause disorganized growth of fibrous and vascular tissues in various parts of the body. The occurrence of these processes at the surface, where the changes can be observed immediately, and without invasive procedures, should have aroused wide interest among those who study kidney disease, diabetes, and other degenerative diseases in which fibrosis and neovascularization play important roles. A localized stress or irritation at first produces vasodilation that increases the delivery of blood to the tissues, allowing them to compensate for the stress by producing more energy. Some of the agents that produce vasodilation also reduce oxygen consumption (nitric oxide, for example), helping to restore a normal oxygen tension to the tissue. Hypoxia itself (produced by factors other than irritation) can induce vasodilation, and if prolonged sufficiently, tends to produce neovascularization and fibrosis. Sensitivity to the harmful effects of light can be increased by some drugs and by excess porphyrins produced in the body (and by the porphyrin precursor, delta-amino levulinic acid), leading to rosacea, so those factors should be considered,

but too often alcohol (which can cause porphyrin to increase) is blamed for rosacea and rhinophyma, without justification. There are many ways in which poor health can increase light sensitivity. Some types of excitation produced by metabolites (or by the failure of inhibitory metabolites) can produce vasodilation, involving the release of nitric oxide (Cardenas, et al., 2000), setting off a series of potentially pathological reactions, including fibrosis. The nitric oxide increases glycolysis while lowering energy production. The excitatory metabolite glutamate, and nitric oxide, are both inhibited by aspirin (Moro, et al., 2000). When blood flow in skin affected by rosacea was measured, circulation was 3 or 4 times higher than normal (Sibenge & Gawkrödger, 1992), and oxygen tension may be increased. An inability to extract oxygen from the blood, or to use it to produce energy, will produce the same hyperemia that would be produced by a lack of oxygen. These measurements suggest that mitochondrial defects would be the best place to look for a general cause of rosacea. When mitochondria are damaged, active cells produce increased amounts of lactic acid, even in the presence of adequate oxygen. Otto Warburg identified this kind of metabolism, aerobic glycolysis, as an essential feature of cancer, and showed that it could be produced by stress, ionizing radiation, carcinogenic toxins, and even by a simple oxygen deficiency. Other investigators around the same time showed that lactic acid produces vasodilation (for example, in the cornea), and more recently it has been shown to promote the development of fibrosis, and it has been called a "phlogogen," a promoter of inflammation. Riboflavin, vitamin B2, is an essential component of the mitochondrial respiratory enzymes, and it is very easily destroyed by light (blue light and especially ultraviolet). When it is excited by high energy light, it can spread the damage to other components of the mitochondria, including the cytochromes and the polyunsaturated fatty acids. The other B vitamins are affected when riboflavin's actions are disturbed. Vitamin K is also extremely light sensitive, and it interacts closely with coenzyme Q in regulating mitochondrial metabolism. For example, mitochondrial Complex-I, NADH-ubiquinone reductase, is probably the most easily damaged part of the mitochondrion, and it is protected by vitamin K. Vitamin E, coenzyme Q, and the polyunsaturated fatty acids are also light sensitive, and they are more susceptible to free radical damage when vitamin K is deficient. Niacinamide, one of the B vitamins, provides energy to this mitochondrial system. Under stress and strong excitation, cells waste niacinamide-NADH, but niacinamide itself has a sedative antiexcitatory effect, and some of its actions resemble a hormone. Estrogen tends to interfere with the formation of niacin from tryptophan. Tryptophan, rather than forming the sedative niacin (pyridine carboxylic acid), can be directed toward formation of the excitatory quinolinic acid (pyridine dicarboxylic acid) by polyunsaturated fatty acids. Excitation must be in balance with a cell's energetic resources, and niacinamide can play multiple protective roles, decreasing excitation, increasing energy production, and stabilizing repair systems. The state of excitation and type of energy metabolism are crucial factors in governing cell functions and survival. The polyunsaturated fatty acids, besides their interactions with estrogen and tryptophan metabolism, promote excitation and decrease energy production in several other ways. For example, they increase the excitatory effects of the glutamate pathways (Yu, et al., 1986; Nishikawa, 1994), and their breakdown products inhibit mitochondrial respiration (Humphries, et al., 1998; Picklo, et al., 1999; Lovell, et al., 2000). The excess excitation that produces nitric oxide and lactic acid lowers the energy production of vascular cells, possibly enough to lower their contractile ability (Geng, et al., 1992), causing vasodilation. When flushing is caused by a mismatch between energy supply and energy demand, caffeine can decrease the vasodilation (Eikvar & Kirkebøen, 1998), but when vasodilation is caused more physiologically by carbon dioxide, caffeine doesn't have that effect (Meno, et al., 2005). In a study in which drinking hot water or coffee was compared with drinking room-temperature coffee or caffeine, it was found that the hot liquids caused flushing, but cool coffee and caffeine didn't. Caffeine increases cells' energy efficiency, and by opposing the effects of adenosine (secreted by cells that are stressed and energy-depleted), it can inhibit vasodilation, angiogenesis (Merighi, et al., 2007; Ryzhov, et al., 2007), and fibrosis (Chan, et al., 2006). One nearly ubiquitous source of inappropriate excitation and energy depletion is the endotoxin, bacterial lipopolysaccharides absorbed from the intestine (Wang and White, 1999). That this ubiquitous toxin has a role in rosacea is suggested by the observation that intestinal stimulation, to speed transit through the bowel, immediately relieved symptoms (Kendall, 2002). Increased cortisol (Simon, et al., 1998) and sepsis (Levy, 2007) interfere with mitochondrial energy production. Simple nervous blushing or flushing is usually considered harmless, and when a person is overheated, the reddening of the skin has the function of facilitating heat loss, to restore a normal temperature. But even nerve-regulated flushing can

involve a distinct interference with mitochondrial respiration, and can stimulate the overgrowth of blood vessels. Cancer's respiratory defect that Warburg identified, fermentation with lactic acid production even in the presence of adequate oxygen, was the result of some kind of injury to the mitochondria. He showed that one of the injuries that could produce aerobic glycolysis was a deficiency of riboflavin. He observed that tumors generally were anoxic, and that cancers typically appeared in the midst of tissue that was atrophying, and suggested that the cancer cells' survival was favored by their ability to live without oxygen. This may be relevant to the observations of many surgeons of a small cancer embedded in the fibrous tissue of large rhinophymas that have been removed. The relatively high incidence of rosacea among women (some studies indicate that it may be 3 times as common in women as in men) isn't likely to be the result of greater sun exposure, so it's reasonable to look for hormonal causes. In old age, it's well recognized that men's estrogen level rises. But the estrogen industry has convinced women that their estrogen declines as they get older. It's common knowledge that aging rodents often go into "persistent estrus," and that their estrogen levels generally increase with age (Parkening, et al., 1978; Anisimov and Okulov, 1981). Several studies in women have shown that serum estrogen levels rise from the teens into the 40s (Musey, et al., 1987; Wilshire, et al., 1995; Santoro, et al., 1996). Other studies show that serum and tissue estrogen concentrations are not concordant, and that some tissues may contain several times as much estrogen as the serum (Jefcoate, et al., 2001). Local irritation increases tissue estrogen content. The antiestrogens, especially progesterone, begin declining in the 30s, so that the rising estrogen has more effect on the tissues during those years. These are the years in which the incidence of rosacea rises suddenly. Rosacea develops later on average in men, whose estrogen levels rise significantly at later ages. Estrogen's most immediate effect on cells is to alter their oxidative metabolism. It promotes the formation of lactic acid. In the long run, it increases the nutritional requirements for the B vitamins, as well as for other vitamins. It also increases the formation of aminolevulinic acid, a precursor of porphyrin, and increases the risk of excess porphyrin increasing light sensitivity. Both aminolevulinic acid and excess porphyrins are toxic to mitochondria, apart from their photosensitizing actions. Nitric oxide, glutamate, and cortisol all tend to be increased by estrogen. Veins and capillaries are highly sensitive to estrogen, and women are more likely than men to have varicose veins, spider veins, leaky capillaries, and other vascular problems besides rosacea. Estrogen can promote angiogenesis by a variety of mechanisms, including nitric oxide (Johnson, et al., 2006). "Estrogens potentiate corticosteroid effects on the skin such as striae, telangiectasiae, and rosacea dermatitis" (Zaun, 1981). Early forms of oral contraceptives, high in estrogen, were found to increase acne rosacea more than three-fold (Prenen & Ledoux-Corbusier, 1971). Lactic acid, produced under the influence of estrogen, nitric oxide, or other problems of energy formation, besides causing vasodilation, also stimulates the growth of fibroblasts. Oxygen deprivation, or damage to mitochondria, will increase lactic acid formation, and so it will immediately cause vasodilation, and if the problem is prolonged, new blood vessels will grow, and fibrous connective tissue will increase. Estrogen stimulates collagen synthesis, and it has been associated with a variety of inflammatory and fibrotic conditions (for example, Cutolo, et al., 2003. Payne, et al., 2006, suggest the use of the anti-estrogen, tamoxifen, to treat rhinophyma.) The cornea normally contains more riboflavin even than the retina, which has a much higher rate of metabolism. When the cornea isn't able to get enough oxygen from the air for its needs (and if riboflavin is deficient, its need for oxygen is increased), surrounding blood vessels at first dilate in response to the diffusing lactic acid, to increase the blood supply to the edges of the cornea. If the problem is prolonged, the conjunctiva becomes chronically blood-shot, hyperemic, and larger more visible blood vessels grow, surrounding the cornea, or even invading the cornea. Many people, especially women, experienced problems of this sort from wearing contact lenses, especially when the lenses were made of materials very impermeable to oxygen (Dumbleton, et al., 2006). Sunlight, and mechanical obstruction of the cornea, produce very localized effects, but those local effects are more likely to be harmful when there is a systemic nutritional deficiency or excess of estrogen. When the systemic problem is very severe, the cheeks, nose, and eyes might not be the first tissues to experience a functional disturbance. The mitochondrial inhibition produced by the action of the parasympathetic nervous system (occurring in simple blushing) can occur wherever those nerves act, and blood vessels in all parts of the body are responsive to the acetylcholine secreted by those nerves. Sleep typically involves a shift of dominance in the autonomic nervous system toward the parasympathetic nerves, with vasodilation. Nosebleeds, especially in children, commonly occur during sleep (Jarjour & Jarjour, 2005: high incidence in sleep, and association with

migraine). A 3 year-old child who had been having an average of 3 nosebleeds every day, during a nap and at night, for several months, also had an extreme behavior problem. He became angry and sometimes violent when he went a little longer than normal between meals. After an oral dose of about ten milligrams of riboflavin, he was able to sleep without having another recurrence of the nosebleeds, and his tantrums became rare. Apparently, the nerve-regulated vasodilation produced by sleep, combined with a riboflavin deficiency, had been enough to produce nosebleeds. The energy deficit resulting from a systemic riboflavin deficiency had probably been causing him to be abnormally sensitive to glycogen depletion, producing sudden anger. In another individual, the energy problem might have taken the form of a memory problem, or of a hemorrhage in the brain or other essential organ. A 37 year old slightly alcoholic man with a bright red nose and cheeks was an amateur fiction writer, but he was having trouble with his memory for words, and for everyday events. Even conversationally, he had to struggle for relatively familiar words. On the suggestion that riboflavin might help his memory, by allowing his brain cells to use oxygen more efficiently, he had his doctor give him an intravenous injection of B vitamins. When I saw him the next day, his conversation was perfectly fluent, and he obviously had easy access to a good vocabulary. Just as noticeable was the normal color of his nose and cheeks. For a week, he had a daily injection of the B vitamins, and his nose color and vocabulary stayed normal. But on the weekend, after not having the shots for two days, his nose and cheeks were again maraschino cherry red, and his speech was halting, as he struggled for words. He forgot the whole episode, and neglected to return to the doctor for more of the vitamin injections. Ten years later, he had developed a medium-sized potato nose, and had his heart valves replaced. His vitamin requirements were apparently abnormally high. At first, the problems resulting from damaged mitochondria seem mostly functional (flushing, mood, memory problems, etc.) and variable, but chronically disturbed functions lead to structural, anatomical changes, as prolonged stimulation alters tissue maintenance and growth. Abram Hoffer, who had been treating schizophrenia and senile dementia with niacin, accidentally discovered that it cured his bleeding gums. That led to its use to treat heart disease. The "orthomolecular" ideas of Hoffer and Linus Pauling were developed in a context of biochemistry governed by genetics, molecular biology, in which the goal was to provide a chemical that was lacking because of a genetic defect in metabolism. Their idea of using nutrients as drugs has led to many unphysiological practices, in which an isolated nutrient is supposed to have a drug-like action, and if in isolation it doesn't act like a drug, then it should be used only according to the normal genetically determined nutritional requirement. But in reality, nutritional requirements are strongly influenced by history and present circumstances. For example, when corneal mitochondria have been damaged by riboflavin deficiency, they have been found to subsequently require more than the normal amount of the vitamin to function properly. And the presence of a certain amount of one nutrient often increases or decreases the amount of other nutrients needed. When the interactions among energy expenditure and energy production, and cellular activation and cellular inhibition, are taken into account, then it's clear that any particular problem is likely to have many causes and many factors that could contribute to a cure. Lactate, glutamate, ammonium, nitric oxide, quinolate, estrogen, histamine, aminolevulinate, porphyrin, ultraviolet light, polyunsaturated fatty acids and endotoxin contribute to excitatory and excitotoxic processes, vasodilation, angiogenesis, and fibrosis. Carbon dioxide, glycine, GABA, saturated fatty acids (for example, Nanji, et al., 1997), vitamin K, coenzyme Q10, niacinamide, magnesium, red light, thyroid hormone, progesterone, testosterone, and pregnenolone are factors that can be increased to protect against inappropriate cellular excitation. All of the nutritional factors that participate in mitochondrial respiration contribute to maintaining a balance between excessive excitation and protective inhibition. Riboflavin, coenzyme Q10, vitamin K, niacinamide, thiamine, and selenium are the nutrients that most directly relate to mitochondrial energy production. Coffee is often avoided by people with rosacea, but it is a very good source of niacin and magnesium, and caffeine has some of the same cell-protective functions as niacinamide. People suffering from rosacea have been found to be more likely than average to have suffered from styes in childhood, to have varicose veins and spider veins, and to suffer from migraines and depression. Hypothyroidism has been identified as a factor in all of those. Good thyroid function is necessary for resistance to bacterial infection, for regulation of blood sugar, neurotransmitters, and hormones related to mood, and for the formation of progesterone. Progesterone regulates smooth muscle tone, including the walls of veins, so that a deficiency allows veins to enlarge. It also prevents overgrowth of fibrotic tissue, and in some contexts may inhibit angiogenesis. GABA itself tends to raise body temperature (Ishiwata, et al., 2005), by controlling vasodilation, and the

factors such as progesterone which protect mitochondrial energy production are also thermogenic, supporting the GABA system. Flushing, both by directly causing heat loss and by reducing mitochondrial energy production, tends to lower body temperature. The sun-damaged areas in rosacea can be directly provided with some of the protective factors by applying them topically. In the same way that topical lactate can cause vasodilation and disturbed energy metabolism (Rendl, et al., 2001), topical niacinamide, progesterone, vitamin K, and coenzyme Q10 can improve the metabolism and function of the local tissues. Riboflavin can probably be useful when applied topically, but because of its extreme sensitivity to light, it should usually be used only internally, unless the treated skin is covered to prevent exposure to light. Topically applied caffeine, even after sun exposure, can reduce local tissue damage (Koo, et al., 2007). Aspirin and saturated fats can also be protective when applied topically. Some of the benefit from antibiotics probably results from the reduced endotoxin stress when intestinal bacteria are suppressed. However, antibiotics can kill the intestinal bacteria that produce vitamin K, so it's important to include that in the diet when antibiotics are used. Some fibers, such as raw carrots, that are effective for lowering endotoxin absorption also contain natural antibiotics, so regular use of carrots should be balanced by occasional supplementation with vitamin K, or by occasionally eating liver or broccoli. Abram Hoffer's research was instrumental in getting niacin recognized as a heart protective drug, but nearly everyone who prescribes it does so to lower blood lipids. That wasn't Hoffer's understanding of its function. He thought it acted directly on blood vessels to protect their integrity. During his studies of its effects on heart disease, he saw that it also lowered cancer mortality, and so began treating cancer patients with it, with considerable success, but there was no medical cliché that could allow the profession to follow in that direction. The arguments I have outlined for considering rosacea to be essentially a problem of metabolic energy, and the mechanisms that I mention for restoring mitochondrial functions, might seem more complex than Hoffer's orthomolecular views. However, this approach is actually much simpler conceptually than any of the ideologies of drug treatment. It simply points out that certain excitatory factors can interfere with energy production, and that there are opposing "inhibitory" factors that can restore energy efficiency. Sometimes, using just one or two of the factors can be curative. Because mitochondrial respiration is very similar in every kind of tissue, a physiological view of rosacea could incline us toward considering the effects of these metabolic factors in other organs during stress and aging--what would the analogous condition of rosacea and rhinophyma be in the brain, heart, liver, or kidney?

REFERENCES

Probl Endokrinol (Mosk), 1981 Mar-Apr, 27:2, 48-52. [Blood estradiol level and G2-chalone content in the vaginal mucosa in rats of different ages] Anisimov VN; Okulov VB "17 beta-Estradiol level was higher in the blood serum of rats aged 14 to 16 months with regular estral cycles during all the phases as compared to that in 3- to 4-month-old female rats." Invest Ophthalmol Vis Sci. 1991 Jun;32(7):1981-5. Analysis of flavins in ocular tissues of the rabbit . Batey DW, Eckhert CD. Riboflavin is the precursor of flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD), coenzymes required for the activity of flavoenzymes involved in the transfer of electrons in oxidation-reduction reactions. Flavins are light sensitive and rapidly degrade when exposed to light in the near ultraviolet and visible wavelengths. Some of the byproducts of flavin photodegradation are toxic. A quantitative survey of flavins in rabbit ocular tissues is reported. Adult male Dutch-Belt Rabbits were fed purified diets containing 3, 30, or 300 mg riboflavin/kg for 1 month. A method of aqueous extraction and high-performance liquid chromatography with fluorescence detection was used to measure riboflavin, FMN, and FAD in cornea, lens cortex, lens nucleus, retina, and blood. The retina contained the highest flavin concentration. In all tissues, the primary flavin was FAD followed by FMN and riboflavin. The highest concentration of riboflavin occurred in the cornea followed by the retina, lens cortex, and lens nucleus. A trend toward increasing concentrations of riboflavin occurred in the retina and blood in response to excess dietary riboflavin, but the concentration changes were not statistically significant. The highest concentration of FAD and FMN occurred in the retina followed by the cornea and the lens cortex and nucleus. The relative contribution of riboflavin, FMN, and FAD to the total flavin pool was markedly different in the various tissues of the eye. The proportion of tissue flavins present as riboflavin decreased from anterior to posterior. It was highest in the cornea followed by lens and retina. The pattern of distribution for FMN was: cornea greater than retina greater than lens cortex and nucleus. Boll Ocul. 1955 Mar;34(3):157-70. [Clinical contribution on riboflavin deficiency of the eye.] [Article in Italian] Bellomio S. Eur J Pharmacol. 2006 Oct 10;547(1-3):184-91. Characterization of the antinociceptive and anti-inflammatory activities of riboflavin in different experimental models. Bertollo CM, Oliveira

AC, Rocha LT, Costa KA, Nascimento EB Jr, Coelho MM. *Vestn Oftalmol.* 1961 Nov-Dec;74:48-53. [The content of riboflavin and ascorbic acid in the cornea in burns of the eye.] [Article in Russian]

Blinova LI, Tsylin LM, Sheinberg AI. *J Neurochem.* 2000 May;74(5):2041-8. Implication of glutamate in the expression of inducible nitric oxide synthase after oxygen and glucose deprivation in rat forebrain slices. Cardenas A, Moro MA, Hurtado O, Leza JC, Lorenzo P, Castrillo A, Bodelon OG, Bosca L, Lizasoain I. *Br J Pharmacol.* 2006 Aug;148(8):1144-55. Adenosine A(2A) receptors play a role in the pathogenesis of hepatic cirrhosis. Chan ES, Montesinos MC, Fernandez P, Desai A, Delano DL, Yee H, Reiss AB, Pillinger MH, Chen JF, Schwarzschild MA, Friedman SL, Cronstein BN. *Clin Exp Rheumatol.* 2003 Nov-Dec;21(6):687-90. New roles for estrogens in rheumatoid arthritis. Cutolo M, Capellino S, Montagna P, Villaggio B, Sulli A, Serio B, Straub RH. Sex hormones appear to play an important role as modulators of autoimmune disease onset/perpetuation. Steroid hormones are implicated in the immune response, with estrogens as enhancers at least of humoral immunity, and androgens and progesterone (and glucocorticoids) as natural immune suppressors. Serum levels of estrogens have been found to be normal in rheumatoid arthritis (RA) patients. Synovial fluid levels (SF) of proinflammatory estrogens relative to androgens are significantly elevated in both male and female RA patients as compared to controls, which is most probably due to an increase in local aromatase activity. Thus, available steroid pre-hormones are rapidly converted to proinflammatory estrogens in the synovial tissue in the presence of inflammatory cytokines (i.e. TNF alpha, IL-1, IL-6). The increased estrogen concentrations observed in RA SF of both sexes are characterized mainly by the hydroxylated forms, in particular 16 alpha-hydroxyestrone, showing a mitogenic stimulating role. Indeed, recent studies by us indicate that 17-beta estradiol (E2) clearly enhanced the expression of markers of cell growth and proliferation, whereas testosterone (T) induced an increase in markers indicating DNA damage and apoptosis. In particular, our data further shows that the enhancing role of estrogens on the immune/inflammatory response is exerted by activating the NFkB complex. In conclusion, locally increased estrogens may exert activating effects on synovial cell proliferation, including macrophages and fibroblasts, suggesting new roles for estrogens in RA. *Tidsskr Nor Laegeforen.* 1998 Mar 30;118(9):1390-5. [Receptor mediated effects of adenosine and caffeine] [Article in Norwegian] Eikvar L, Kirkebøen KA. "Adenosine consists of one ribose and one purine moiety and binds to specific receptors on cell membranes. The receptors are coupled to G-proteins and additionally to various effector-systems. When a mismatch occurs between energy supply and energy demand, adenosine is produced by the catabolism of adenosine triphosphate. The metabolism of an organ is thereby coupled to the local blood supply (metabolic vasodilation)." *Arch Dermatol.* 1988 Jan;124(1):72-9. Health effects of sunlight exposure in the United States. Results from the first National Health and Nutrition Examination Survey, 1971-1974. Engel A, Johnson ML, Haynes SG. The dermatologic component of the first National Health and Nutrition Examination Survey (N = 20,637), conducted from 1971 through 1974, documented the deleterious effect of ultraviolet radiation on selected skin and eye conditions. Actinic skin damage was more frequent in white men with high as compared with low sunlight exposure, 36.7% vs 23.3%, respectively. Among white women, the corresponding figures were 34.1% vs 18.6%, respectively. Actinic damage was found more often in individuals with light eye color. Basal cell epitheliomas were found in 11.3% of white men aged 65 to 74 years who had severely actinic-damaged skin as compared with 1.0% of those with undamaged skin. Sunlight exposure was positively associated with localized hypomelanism, localized hypermelanism, seborrheic keratoses, senile lentigines, freckles, acne rosacea, spider nevi, varicose veins, venus star, dry skin, wrinkled skin, pterygia, arcus senilis, and a variety of minor oral lesions of the tongue, palate, and buccal mucosa. These findings suggest that a large number of dermatologic conditions, which may in part result from overexposure to sunlight, may be preventable. *Nippon Ganka Gakkai Zasshi.* 1961 Dec 10;65:2439-44. [The effects of vitamin B2 group on the corneal metabolism. I.] [Article in Japanese] Funatsu H, Motegi T. *J Cosmet Dermatol.* 2004 Apr;3(2):88-93. Nicotinic acid/niacinamide and the skin. Gehring W. *Circ Res.* 1992 Nov;71(5):1268-76. Interferon-gamma and tumor necrosis factor synergize to induce nitric oxide production and inhibit mitochondrial respiration in vascular smooth muscle cells. Geng Y, Hansson GK, Holme E. "Nitric oxide (NO) is an important signal substance in cell-cell communication and can induce relaxation of blood vessels by activating guanylate cyclase in smooth muscle cells (SMCs)." "It was recently shown that SMCs may themselves produce NO or an NO-related compound. We have studied NO production and its effects on energy metabolism in cultured rat aortic smooth muscle cells. It was observed that the cytokines, interferon-gamma and

tumor necrosis factor- α , synergistically induced an arginine-dependent production of NO in these cells. This was associated with an inhibition of complex I (NADH: ubiquinone oxidoreductase) and complex II (succinate: ubiquinone oxidoreductase) activities of the mitochondrial respiratory chain, suggesting that NO blocks mitochondrial respiration in these cells. Lactate accumulated in the media of the cells, implying an increased anaerobic glycolysis, but there was no reduction of viability. An NO-dependent inhibition of mitochondrial respiration and a switch to anaerobic glycolysis would reduce energy production of the SMCs. This would in turn reduce the contractile capacity of the cell and might represent another NO-dependent vasodilatory mechanism." *Q Bull Northwest Univ Med Sch.* 1952;26(2):120-3. Riboflavin and the cornea. Gordon OE. *Bull Soc Fr Dermatol Syphiligr.* 1950 May-Jun;57(3):277-80. Cutaneous-mucosal ariboflavinosis; rosacea of cornea and medio-facial seborrheic dermatitis. Gougerot H, Grupper C, Plas G. *Br J Dermatol.* 2005 Dec;153(6):1176-81. Comorbidity of rosacea and depression: an analysis of the National Ambulatory Medical Care Survey and National Hospital Ambulatory Care Survey--Outpatient Department data collected by the U.S. National Center for Health Statistics from 1995 to 2002. Gupta MA, Gupta AK, Chen SJ, Johnson AM. *Biochemistry.* 1998 Nov 10;37(45):15835-41. Selective inactivation of alpha-ketoglutarate dehydrogenase and pyruvate dehydrogenase: reaction of lipoic acid with 4-hydroxy-2-nonenal. Humphries KM, Szweda LI. "Previous research has established that 4-hydroxy-2-nonenal (HNE), a highly toxic product of lipid peroxidation, is a potent inhibitor of mitochondrial respiration. HNE exerts its effects on respiration by inhibiting alpha-ketoglutarate dehydrogenase (KGDH). Because of the central role of KGDH in metabolism and emerging evidence that free radicals contribute to mitochondrial dysfunction associated with numerous diseases, it is of great interest to further characterize the mechanism of inhibition." "These results therefore identify a potential mechanism whereby free radical production and subsequent lipid peroxidation lead to specific modification of KGDH and PDH and inhibition of NADH-linked mitochondrial respiration." *Biochemistry.* 1998 Jan 13;37(2):552-7. Inhibition of NADH-linked mitochondrial respiration by 4-hydroxy-2-nonenal. Humphries KM, Yoo Y, Szweda LI. During the progression of certain degenerative conditions, including myocardial ischemia-reperfusion injury, mitochondria are a source of increased free-radical generation and exhibit declines in respiratory function(s). It has therefore been suggested that oxidative damage to mitochondrial components plays a critical role in the pathology of these processes. Polyunsaturated fatty acids of membrane lipids are prime molecular targets of free-radical damage. A major product of lipid peroxidation, 4-hydroxy-2-nonenal (HNE), is highly cytotoxic and can readily react with and damage protein. In this study, the effects of HNE on intact cardiac mitochondria were investigated to gain insight into potential mechanisms by which free radicals mediate mitochondrial dysfunction. Exposure of mitochondria to micromolar concentrations of HNE caused rapid declines in NADH-linked but not succinate-linked state 3 and uncoupled respiration. The activity of complex I was unaffected by HNE under the conditions of our experiments. Loss of respiratory activity reflected the inability of HNE-treated mitochondria to meet NADH demand during maximum rates of O₂ consumption. HNE exerted its effects on intact mitochondria by inactivating alpha-ketoglutarate dehydrogenase. These results therefore identify a potentially important mechanism by which free radicals bring about declines in mitochondrial respiration. *Tohoku J Exp Med.* 1954 Dec 25;61(1):93-104. Contribution to the ocular manifestation of riboflavin deficiency. Irinoda K, Sato S. *Brain Res.* 2005 Jun 28;1048(1-2):32-40. Changes of body temperature and thermoregulatory responses of freely moving rats during GABAergic pharmacological stimulation to the preoptic area and anterior hypothalamus in several ambient temperatures. Ishiwata T, Saito T, Hasegawa H, Yazawa T, Kotani Y, Otokawa M, Aihara Y. *Pediatr Neurol.* 2005 Aug;33(2):94-7. Migraine and recurrent epistaxis in children. Jarjour IT, Jarjour LK. *J Natl Cancer Inst Monogr* 2000;(27):95-112. Tissue-specific synthesis and oxidative metabolism of estrogens. Jefcoate CR, Liehr JG, Santen RJ, Sutter TR, Yager JD, Yue W, Santner SJ, Tekmal R, Demers L, Pauley R, Naftolin F, Mor G, Berstein L "However, breast cancer tissue E2 levels are 10-fold to 50-fold higher in postmenopausal women than predicted from plasma levels." *Endocrine.* 2006 Dec;30(3):333-42. Effects of estradiol-17 β on expression of mRNA for seven angiogenic factors and their receptors in the endometrium of ovariectomized (OVX) ewes. Johnson ML, Grazul-Bilska AT, Redmer DA, Reynolds LP. *Clin Exp Dermatol.* 2004 May;29(3):297-9. Remission of rosacea induced by reduction of gut transit time. Kendall SN. Rosacea is a chronic disorder characterized by hypersensitivity of the facial vasculature, presenting with intense flushing eventually leading to chronic erythema and telangiectasia. Although the precise aetiology of rosacea is not known,

numerous associations with inflammatory gastrointestinal tract disorders have been reported. Furthermore, substance P-immunoreactive neurones occur in considerably greater numbers in tissue surrounding affected blood vessels suggesting involvement of neurogenic inflammation and moreover plasma kallikrein-kinin activation is consistently found in patients. In this report, a patient without digestive tract disease is described, who experienced complete remission of rosacea symptoms following ingestion of a material intended to sweep through the digestive tract and reduce transit time below 30 h. It is possible that intestinal bacteria are capable of plasma kallikrein-kinin activation and that flushing symptoms and the development of other characteristic features of rosacea result from frequent episodes of neurogenic inflammation caused by bradykinin-induced hypersensitization of facial afferent neurones. The possible relevance of this hypothesis to other conditions featuring afferent hypersensitivity, such as fibromyalgia, is considered. *Br J Dermatol.* 2007 May;156(5):957-64. Protection from photodamage by topical application of caffeine after ultraviolet irradiation. Koo SW, Hirakawa S, Fujii S, Kawasumi M, Nghiem P. *Medicina (B Aires)* 1985;45(2):110-6. [Fibrosis and cirrhosis in the rabbit induced by diethylstilbestrol and its inhibition with progesterone]. [Article in Spanish] Lanari A, de Kremer GH. *Free Radic Biol Med.* 2000 Oct 15;29(8):714-20. Acrolein, a product of lipid peroxidation, inhibits glucose and glutamate uptake in primary neuronal cultures. Lovell MA, Xie C, Markesbery WR. *J Cereb Blood Flow Metab.* 2005 Jun;25(6):775-84. Effect of caffeine on cerebral blood flow response to somatosensory stimulation. Meno JR, Nguyen TS, Jensen EM, Alexander West G, Groysman L, Kung DK, Ngai AC, Britz GW, Winn HR. "Hypercarbic vasodilatation was unaffected by either caffeine or theophylline." *Mol Pharmacol.* 2007 Aug;72(2):395-406. Caffeine inhibits adenosine-induced accumulation of hypoxia-inducible factor-1alpha, vascular endothelial growth factor, and interleukin-8 expression in hypoxic human colon cancer cells. Merighi S, Benini A, Mirandola P, Gessi S, Varani K, Simioni C, Leung E, MacLennan S, Baraldi PG, Borea PA. *Neuropharmacology.* 2000 Apr 27;39(7):1309-18. Mechanisms of the neuroprotective effect of aspirin after oxygen and glucose deprivation in rat forebrain slices. Moro MA, De Alba J, Cardenas A, De Cristobal J, Leza JC, Lizasoain I, Diaz-Guerra MJ, Bosca L, Lorenzo P. *Am J Obstet Gynecol* 1987 Aug;157(2):312-317. Age-related changes in the female hormonal environment during reproductive life. Musey VC, Collins DC, Musey PI, Martino-Saltzman D, Preedy JR "We found that increased age during reproductive life is accompanied by a significant rise in both basal and stimulated serum follicle-stimulating hormone levels. This was accompanied by an increase in the serum level of estradiol-17 beta and the urine levels of estradiol-17 beta and 17 beta-estradiol-17-glucosiduronate." "Serum levels of dehydroepiandrosterone and dehydroepiandrosterone sulfate decreased with age, but serum testosterone was unchanged." *Hepatology.* 1997 Dec;26(6):1538-45. Dietary saturated fatty acids down-regulate cyclooxygenase-2 and tumor necrosis factor alpha and reverse fibrosis in alcohol-induced liver disease in the rat. Nanji AA, Zakim D, Rahemtulla A, Daly T, Miao L, Zhao S, Khwaja S, Tahan SR, Dannenberg AJ. *J Physiol.* 1994 Feb 15;475(1):83-93. Facilitatory effect of docosahexaenoic acid on N-methyl-D-aspartate response in pyramidal neurones of rat cerebral cortex. Nishikawa M, Kimura S, Akaike N. *J Gerontol.* 1978 Mar, 33:2, 191-6. Circulating plasma levels of pregnenolone, progesterone, estrogen, luteinizing hormone, and follicle stimulating hormone in young and aged C57BL/6 mice during various stages of pregnancy. Parkening TA; Lau IF; Saksena SK; Chang MC. *Antioxid Redox Signal.* 1999 Fall;1(3):255-84. 4-Hydroxynonenal as a biological signal: molecular basis and pathophysiological implications. Parola M, Bellomo G, Robino G, Barrera G, Dianzani MU. *Ann Plast Surg.* 2002 Jun;48(6):641-5. Further evidence for the role of fibrosis in the pathobiology of rhinophyma. Payne WG, Wang X, Walusimbi M, Ko F, Wright TE, Robson MC. *Ann Plast Surg.* 2006 Mar;56(3):301-5. Down-regulating causes of fibrosis with tamoxifen: a possible cellular/molecular approach to treat rhinophyma. Payne WG, Ko F, Anspaugh S, Wheeler CK, Wright TE, Robson MC. *Neurochem Res.* 1991 Sep;16(9):983-9. Release of arachidonic acid by NMDA-receptor activation in the rat hippocampus. Pellerin L, Wolfe LS. *J Neurochem.* 1999 Apr;72(4):1617-24. 4-Hydroxy-2(E)-nonenal inhibits CNS mitochondrial respiration at multiple sites. Picklo MJ, Amarnath V, McIntyre JO, Graham DG, Montine TJ. *Biochim Biophys Acta.* 2001 Feb 14;1535(2):145-52. Acrolein inhibits respiration in isolated brain mitochondria. Picklo MJ, Montine TJ. *Arch Belg Dermatol Syphiligr.* 1971 Jul-Sep;27(3):253-8. [Hormonal contraception and dermatology] Prenen M, Ledoux-Corbusier M. PIP: A group of 150 Belgian women aged 17-47 taking various oral contraceptives was compared with a control group of 400 with respect to incidence of cholasma and piloseborrhic disorders. While taking pills, 10% of alopecia and 5.3% of

juvenile acne cases were improved, but the following symptoms appeared: hypertrichosis in 4%, alopecia in 14.6, juvenile acne in 16.6%, and rosaceous acne in 3.3%. The incidence of these symptoms in the controls was cholasma .75%, hypertrichosis 5.25%, alopecia 11.5%, juvenile acne 21%, and acne rosacea 1%. The pill seemed to aggravate cholasma and rosaceous acne or to improve piloseborrhic symptoms, depending on whether the formulation was dominant in estrogen or progestagen. *Ann Plast Surg.* 2000 Nov;45(5):515-9. Overexpression of transforming growth factor beta-2 and its receptor in rhinophyma: an alternative mechanism of pathobiology. Pu LL, Smith PD, Payne WG, Kuhn MA, Wang X, Ko F, Robson MC. "These findings support the authors' hypothesis that fibrosis may also play an important role in the pathobiology of rhinophyma." www.emedicine.com/oph/topic115.htm Last Updated: May 14, 2007. Author: J Bradley Randleman, MD, Assistant Professor, Department of Ophthalmology, Cornea, External Disease, and Refractive Surgery Section, Emory University School of Medicine *Br J Dermatol.* 2001 Jul;145(1):3-9. Topically applied lactic acid increases spontaneous secretion of vascular endothelial growth factor by human reconstructed epidermis. Rendl M, Mayer C, Weninger W, Tschachler E. *Nature.* 1969 Aug 2;223(5205):516-7. Diminished responsiveness to thyroid hormone in riboflavin-deficient rats. Rivlin RS, Wolf G. *J Pharmacol Exp Ther.* 2007 Feb;320(2):565-72. Role of adenosine receptors in the regulation of angiogenic factors and neovascularization in hypoxia. Ryzhov S, McCaleb JL, Goldstein AE, Biaggioni I, Feoktistov I. *J Clin Endocrinol Metab* 1996 Apr;81(4):1495-501, Characterization of reproductive hormonal dynamics in the perimenopause. Santoro N, Brown JR, Adel T, Skurnick JH "Overall mean estrone conjugate excretion was greater in the perimenopausal women compared to that in the younger women [76.9 ng/mg Cr (range, 13.1-135) vs. 40.7 ng/mg Cr (range, 22.8-60.3); P = 0.023] and was similarly elevated in both follicular and luteal phases." "We conclude that altered ovarian function in the perimenopause can be observed as early as age 43 yr and include hyperestrogenism, hypergonado-tropism, and decreased luteal phase progesterone excretion. These hormonal alterations may well be responsible for the increased gynecological morbidity that characterizes this period of life." *J Am Acad Dermatol.* 1992 Apr;26(4):590-3. Rosacea: a study of clinical patterns, blood flow, and the role of Demodex folliculorum. Sibenge S, Gawkrödger DJ. *Cesk Oftalmol.* 1951;7(1):37-42. [Effect of vitamin B1 and B2 on diseases of the cornea.] Simkova M. *FEBS Lett.* 1998 Sep 11;435(1):25-8. Glucocorticoids decrease cytochrome c oxidase activity of isolated rat kidney mitochondria. Simon N, Jolliet P, Morin C, Zini R, Urien S, Tillement JP. "The importance of mitochondria is rising as a target in pathologic processes such as ischemia." "A regulation of cytochrome c oxidase activity by glucocorticoids will be of particular interest in pathology involving metabolic insult." *Am J Ophthalmol.* 1950 Jul;33(7):1127-36. The ocular manifestations of riboflavin deficiency. Stern JJ. *Arch Ophthalmol.* 1949 Oct;42(4):438-42. Conditioned corneal vascularity in riboflavin deficiency; report of a case. Stern HJ. *Nippon Ganka Kiyo.* 1962 Nov;13:489-94. [Variations of total vitamin B2 content in the cornea, iris and ciliary body and the blood of rabbits in stress. (A preliminary report)] [Article in Japanese] Taketani T. *Free Radic Res.* 1996 Jun;24(6):421-7. The effect of the lipid peroxidation product 4-hydroxynonenal and of its metabolite 4-hydroxynonenic acid on respiration of rat kidney cortex mitochondria. Ullrich O, Henke W, Grune T, Siems WG. *J Neurochem.* 1999 Feb;72(2):652-60. The bacterial endotoxin lipopolysaccharide causes rapid inappropriate excitation in rat cortex. Wang YS, White TD. *Ophthalmologica.* 1955 Jun;129(6):389-95. [Corneal changes in riboflavin-deficient rats.] [Article in German] Wiesinger H, Kaunitz H, Slanetz CA. *Arzneimittelforschung* 1968 Dec;18(12): 1525-9. On the phlogogenic properties of lactic acid in animal experiments. Wilhelmi G, Gdynia R. *Ann Intern Med* 1981 Oct;95(4):468-76. Flushing reactions: consequences and mechanisms. Wilkin JK *J Clin Endocrinol Metab* 1995 Feb;80(2):608-613. Diminished function of the somatotrophic axis in older reproductive-aged women. Wilshire GB, Loughlin JS, Brown JR, Adel TE, Santoro N "Studies in aging women have rarely controlled for menstrual cycle stage or status or body mass index. We hypothesized that GH and IGF-I levels in reproductive-aged women fall with age despite the stimulatory effects of endogenous estradiol (E2). Eight older reproductive-aged women (aged 42-46 yr) with regular menses, of normal weight, and in good health were compared to a group of eight young control subjects (aged 19-34 yr)." "Older reproductive-aged women had lower 12-h integrated daytime GH concentrations (mean +/- SE, 171 +/- 35 vs. 427 +/- 130 micrograms min/L; P = 0.036) than younger controls and a strong trend for lower IGF-I levels (22.7 +/- 2.1 vs. 31.3 +/- 3.5 nmol/L; P = 0.055) than younger controls despite having higher circulating E2 on the day of sampling (368 +/- 51 vs. 167 +/- 20 pmol/L; P = 0.002). We conclude that older reproductive-aged women have lower

daytime GH concentrations than younger controls despite having higher E2 levels on the day of sampling and overall normal gonadal hormone parameters." J Neurochem 1986 Oct;47(4):1181-9.

Effects of arachidonic acid on glutamate and gamma-aminobutyric acid uptake in primary cultures of rat cerebral cortical astrocytes and neurons. Yu AC, Chan PH, Fishman RA. Med Monatsschr Pharm. 1981 Jun;4(6):161-5. [Skin changes from taking hormonal contraceptives] [Article in German] Zaun H. PIP: Hormonal contraceptives can induce changes in the skin and its appendages. "Estrogens potentiate corticosteroid effects on the skin such as striae, telangiectasiae, and rosacea dermatitis." © Ray Peat Ph.D. 2013. All Rights Reserved.

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